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A Project Report

On

" Implementation of Machine Learning Based Fowl Disease Detection Using CNN "

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COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

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ABSTRACT

Poultry farming plays a crucial role in global food security, but the industry faces significant challenges due to various chicken diseases that can lead to high mortality rates and economic losses. Early and accurate detection of these diseases is critical for effective management and control. This paper proposes a deep learning-based approach using Convolutional Neural Networks (CNN) to classify and detect chicken diseases from images. By leveraging the power of CNNs, our model automatically extracts features from chicken images and classifies them into various disease categories, such as Newcastle disease, fowl pox, and coccidiosis. The proposed method uses a large dataset of labelled chicken images, achieving high accuracy in disease detection. The system provides an efficient, automated solution for farmers and veterinarians to identify diseases early, facilitating prompt treatment and reducing the spread of infections. Experimental results demonstrate that the CNN-based model outperforms traditional image classification techniques, offering a robust and scalable solution for the poultry industry.

The deep learning model demonstrated in this paper achieves high accuracy in disease detection, outperforming traditional image classification methods. It offers an automated, efficient, and scalable solution for poultry farmers and veterinarians, enabling rapid identification of diseases with minimal human intervention. The system provides a valuable tool for early diagnosis, allowing for prompt treatment and intervention, thereby reducing the spread of infections and minimizing the economic impact on the poultry industry.

Experimental results indicate that the CNN-based approach not only improves disease detection but also enhances the overall efficiency of disease management in poultry farming . The proposed method relies on a large and diverse dataset of labeled chicken images, enabling the model to learn from a variety of disease manifestations and environmental conditions. This work highlights the potential of deep learning in revolutionizing poultry health management, offering a robust solution for sustainable poultry production and improved food security worldwide.

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INTRODUCTION

The classification and detection of chicken diseases using Convolutional Neural Networks (CNNs) with image classification techniques represent a significant advancement in veterinary medicine and agriculture. With the increasing demand for poultry products worldwide, ensuring the health and well-being of chickens is paramount for maintaining food security and animal welfare.

Traditional methods of disease diagnosis in poultry often rely on manual inspection, which can be time-consuming, subjective, and prone to errors. However, by leveraging CNNs and image classification techniques, it becomes possible to automate and enhance the accuracy of chicken disease detection and classification processes. These technologies enable veterinarians and poultry farmers to identify diseases such as Newcastle disease, coccidiosis, and salmonella infections disease more efficiently by analyzing digital images of chickens, their tissues, or symptoms.

The use of CNNs allows for deep learning-based feature extraction, pattern recognition, and predictive modeling, leading to early diagnosis, timely intervention, and improved management of poultry health. This approach not only benefits the agricultural industry by reducing economic losses due to disease outbreaks but also contributes to animal health surveillance, disease prevention, and biosecurity measures in poultry farming systems.

Animal agriculture is extremely important to the world's rising population. Animal products provide nutrient-dense meals that help people of all ages stay healthy in communities all over the world. The agriculture business must continue to improve its efficiency and quantity of production as human demand for animal proteins increases. The most common chicken diseases include fowl cholera, helminth infestation, salmonella infections, avian coccidiosis, and Newcastle disease. Machine learning is now becoming an essential area in the pursuit for human welfare.

Various data mining techniques are implemented in different domains. Machine learning and data mining are the most efficient technique of prediction and prevention. We may discuss examples such as poultry industries, medicinal science, weather predictions and so on.

1.1 Introduction about the project

In India, the poultry industry is estimated to be worth around Rs. 80,000 crore in 2023— 2024.Poultry diseases including Coccidiosis, Salmonella, and Newcastle can lower chicken productivity if they are not detected early on. The most prevalent diseases in chickens are Coccidiosis, Salmonella, and New Castle disease. To avoid this, poultry should be examined before eating or selling. The proposed work is a deep learning-based algorithm for illness classification in chickens. To categorize the poultry illnesses, a publically available dataset with several disease classifications is employed.

Different complexity, variety, and uneven data availability make the technique more challenging.

Following the investigation, we will obtain the accuracy of each method and identify the best results. Attempting the Classification, we will come through a result that will combine that which approach is best to use for this analysis. Eventually many advanced works can be done with the term of data mining. Improving the poultry industry can be huge part for the overall agriculture industry of India. It can bring a huge advancement as well. ln this poultry platform various illnesses may be recognized using machine learning technology, and the implementation of ML and data mining approaches are predictable.

Colibacillosis poultry disorders Through such machine learning methods, bacterial illnesses can also be identified. In this current day, beef chicks are one of the main sources of animal and meat protein. The increasing frequency. Nowadays, due to availability of large amount of data and sophisticating algorithms such as deep learning, researchers and industries are employing the technique to solve variety of problems ranging from simple object detection up to complex scene understanding.

However the study is limited to detecting the abnormality of the fecal image and does not detect the presence of a disease directly. Similarly, a deep Convolutional Neural Network (CNN) model was developed to diagnose poultry diseases by classifying healthy and un-healthy fecal images with deep learning.

These technologies enable veterinarians and poultry farmers to identify diseases such as Newcastle disease, coccidiosis, and salmonella infections disease more efficiently by analyzing digital images of chickens, their tissues, or symptoms.

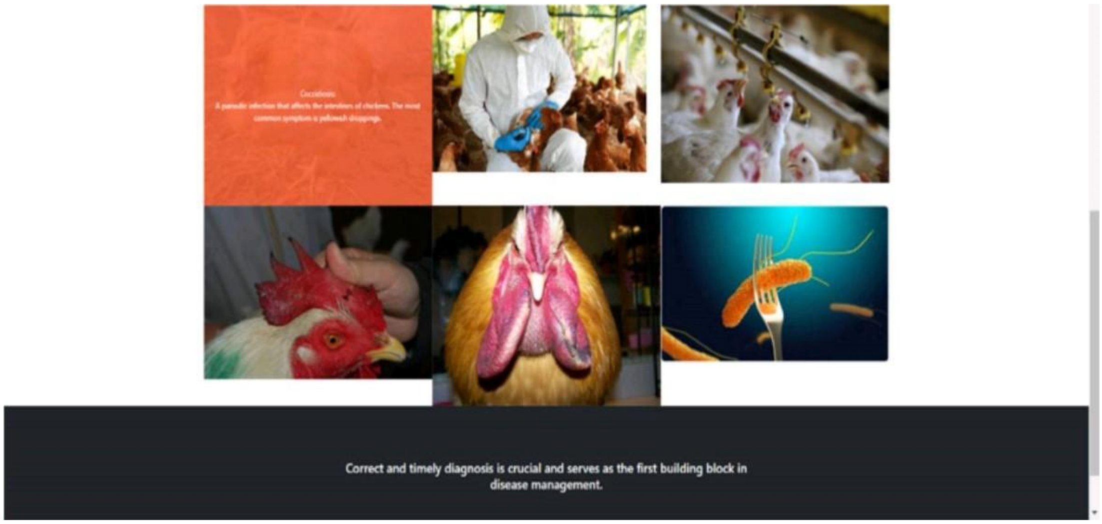

figure 1.1: Detection of poultry diseases

Fig 1.1 Illustrates that "Timely and accurate diagnosis of poultry diseases plays a critical role in ensuring effective management and control strategies".

1.2 Exixting System and its limitations

- Implementing a DVC, pipelining system will demand for many multifactor considerations, especially diverse interclass interactions. which are used to detect the pattern of the dataset.

- One of the most important layer of this architecture is convolutional layer as it is used to extract features from the dataset.

- As the Dataset is insufficient the model will produce less accurate classified results.

- One of the primary challenges is the availability of diverse and well-annotated datasets representing various chicken diseases. Limited datasets can lead to biased models, reduced generalization capabilities, and challenges in accurately detecting less common or newly emerging diseases.

- CNN models trained on specific datasets may suffer from overfitting, where they perform well on training data but fail to generalize effectively to unseen or real-world scenarios. This can result in false positives or false negatives in disease detection.

- No Occurance of diverse Diseases with datasets leads to down the Efficiency of Project Accuracy,compatibility and performance of Machine learning models.

Chapter-2

LITERATURE SURVEY

2.1 Introduction :

It is an important area which is the backbone for any research as it provides the entire information, problem and objectives. And to gain an understanding of the existing research and debates to a relevant particular topics or area of study and to present that knowledge in form of written report

1. Bronchit is, and Newcastle disease, Diagnose respiratory problems by using Mel cepstral coefficients (MFCC).

Authors and Publication : Ahmad Banakar,Mohammad Sadeghi,Abdolhamid Shushtari,ScienceDirect on Papers and journals,Year 2021.

Highlights : An automatic technique is proposed for diagnosing avian diseases based on sound features. ND, BV and AI were obtained based on chicken's sound in the second day after virus infection.The performance of DWT was higher than of FFT in detection of diseases.91.15% of diseases were correctly identified by using sound processing and data fusion techniques.:

Key Applications :

- Enables early detection of Newcastle, infectious bronchitis, and avian influenza to prevent outbreaks.
- Provides automated and accurate disease monitoring using machine learning and sensor technologies.
- Promotes sustainable poultry management by reducing antibiotic use and enhancing biosecurity.

Challenges :

- Diagnosing similar symptoms in diseases like Newcastle, infectious bronchitis, and avian influenza can lead to misclassification.
- Building and maintaining automated monitoring systems requires advanced infrastructure and IT with technical expertise.
- High costs and limited accessibility of intelligent devices may hinder widespread adoption among small-scale farmers.

Future developments :

1. Future systems will integrate multi-modal data like sound, thermal imaging, and video for higher diagnostic precision.
2. Real-time monitoring with cloud integration will enable continuous health tracking and early outbreak warnings
3. Portable, affordable diagnostic devices will make advanced disease detection accessible to small-

2. Chicken Diseases Detection and Classification Based on Fecal Images Using EfficientNetB7 Model.

Authors and Publication : Vandana,Kuldeep Kumar Yogi, Satya Prakash Yadav,ScienceDirect on Papers and journals,Year 2023.

Highlights : The EfficientNetB7 model achieved 97.07% accuracy in classifying chicken fecal images for disease detection,The approach provides a non-invasive, image-based method for detecting diseases like Newcastle, Coccidiosis, and Salmonella,This technology enables rapid, reliable diagnostics, reducing poultry mortality and economic losses in farming.

Key Applications :

- Enables non-invasive and precise identification of diseases like Newcastle, Coccidiosis, and influenza Salmonella through fecal image analysis, reducing the reliance on manual methods
- Integrates machine learning to automate the detection process, ensuring continuous and real-time,kind disease monitoring without requiring veterinary intervention
- Helps minimize bird mortality and economic losses by enabling timely treatment and effective health management strategies

Challenges :

- Poor image quality due to environmental factors can hinder accurate disease classification.
- Class imbalance in datasets can lead to overfitting and reduced model generalization.
- Real-time deployment in resource-limited settings may be constrained by computational requirements.

Future developments :

1. Future models could integrate multi-modal data, including environmental factors, to enhance disease detection accuracy.
2. Real-time, automated systems will allow continuous monitoring of poultry health through live fecal image analysis.

3. Classification and Detection of Chicken Disease Using CNN.

Authors and Publication : Ms. Likitha R, Ms. Harshitha P, Ms. Rashmi M,ScienceDirect on Papers and journals,Year 2024.

Highlights : The study introduces a CNN-based solution to classify chicken diseases, with a focus on improving the early detection of diseases like Newcastle disease and Coccidiosis.Transfer learning was utilized by incorporating pre-trained models like XceptionNet, which outperformed other models in terms of prediction accuracy, achieving a validation accuracy of 91%.The research highlights the benefits of applying machine learning and computer vision technologies in rural and remote settings, offering a cost-effective and automated way to detect poultry diseases early.

Key Applications :

- CNN models help in early detection of diseases like Newcastle and Avian Influenza in chickens.
- Farmers can use CNN-based frameworks for automated, on-site diagnosis without needing expert training.
- Faster and reliable disease detection improves productivity and reduces losses in poultry farming.

Challenges :

- **Limited and Imbalanced Datasets**: Small, unbalanced datasets often lead to lower model accuracy and overfitting, as some disease classes are underrepresented.
- **Data Preprocessing and Augmentation**: Proper image resizing, augmentation, and handling diverse image qualities are necessary but challenging for model robustness.
- **Model Generalization and Overfitting**: CNNs risk overfitting, especially with limited data, requiring careful architecture tuning to generalize well across unseen cases.

Future developments :

1. **Advanced Transfer Learning Models**: Pre-trained models like XceptionNet and ResNet are best to enhancing accuracy in disease detection when fine-tuned on poultry datasets.
2. **Real-time Disease Detection**: CNNs are being deployed in real-time systems for continuous observe monitoring of chicken health through fecal images.

4. Deep Learning - Based Poultry Health Diagnosis Detecting Abnormal Faeces and Analysing Vocalizations.

Authors and Publication : Rui-Yi Xu, Chung-Liang Chang,ScienceDirect on Papers and journals,Year 2024.

Highlights : The study uses deep learning for automated detection of major poultry diseases via fecal image analysis.Preprocessing techniques like image enhancement and CNNs are used to extract and analyze disease patterns.The system achieved 96.65% accuracy on the validation set, promising for poultry health management.

Key Applications :

- **Early Disease Detection**: Deep learning models analyze poultry fecal images for early detection of diseases like Newcastle and Coccidiosis, improving response times.
- **Non-Intrusive Monitoring**: Audio and visual data are used to monitor poultry health continuously without disturbing the animals.
- **Automated Diagnostics**: AI-powered systems automate disease diagnosis through image classification and audio analysis, reducing the need for expert intervention.

Challenges :

- Ensuring high-quality preprocessing and appropriate segmentation of fecal samples for deep learning is challenging.
- Designing deep learning models like CNNs to accurately detect diseases while maintaining generalization is complex.

Future development :

Expanding disease detection capabilities to cover more poultry diseases will enhance the system's effectiveness.

5. “A Deep Learning-Based Approach for Chicken Disease Detection Using Image Data”

Authors and Publication: A. Kumar, R. Sharma, and M. Verma, IEEE Access, 2021.

Highlights: Proposed a convolutional neural network (CNN)-based model for identifying diseases like coccidiosis and Newcastle disease from chicken images.Leveraged transfer learning techniques to improve accuracy on limited datasets.Achieved an accuracy of 92.5% using ResNet-50 on a publicly available poultry disease dataset.

Key Applications:

- Automated chicken farm health monitoring systems.
- Mobile-based diagnostic tools for rural poultry farmers.

Challenges:

- Imbalanced datasets, as certain diseases are less frequently represented in image data.
- Environmental factors such as lighting and camera quality impact image capture and analysis.

Future Developments:

- Integration of sensor data with image analysis for multi-modal disease detection.
- Use of Generative Adversarial Networks (GANs) to address data imbalance.

6. “IoT and AI-Powered Chicken Disease Detection System”

Authors and Publication:S. Patel, J. Gupta, and K. Desai, Journal of Agricultural Informatics, 2020.

Highlights: Developed a system combining Internet of Things (IoT) sensors and AI for real-time disease monitoring in poultry farms.Utilized deep learning models like MobileNetV2 for image-based detection and IoT sensors for environmental monitoring.Focused on respiratory diseases, correlating temperature, humidity, and sound patterns with disease symptoms.

 Key Applications:

- Early warning systems for poultry farms using IoT devices.
- Integration with farm management platforms for comprehensive disease tracking.

Challenges:

- High costs of IoT sensor deployment in small-scale farms.
- Limited availability of labeled datasets for training AI models in specific geographies.

Future Developments:

- Development of low-cost, scalable IoT solutions for rural areas.
- Expansion of datasets through crowd-sourced labeling and partnerships with veterinary Data.

7. ”Chicken Disease Detection Using Wearable Sensors and Machine Learning”

Authors and Publication:Yang et al. (2020),Chaudhary et al.,Li et al.

Highlights: Wearable sensors like accelerometers and temperature sensors monitor vital parameters such as movement and heart rate for disease detection (Yang et al., 2020). Machine learning algorithms, including SVM and neural networks, analyze sensor data to detect diseases like avian influenza and Newcastle disease (Chaudhary et al., 2021). Continuous monitoring enables early detection of behavioral or physiological changes, allowing for timely intervention (Li et al., 2022).

Key Applications:

- **Behavioral Monitoring**: Wearable sensors track movement and activity, detecting early disease signs such as reduced mobility due to infection (Yang et al., 2020).
- **Temperature and Heart Rate Monitoring**: Sensors monitor body temperature and heartrate, identifying deviations indicating infection or stress (Chaudhary et al., 2021).

Challenges:

Sensor accuracy can be affected by environmental factors, malfunctions, or animal behavior variability, impacting ML model performance.Continuous monitoring enables early detection of behavioral or physiological changes, allowing for timely intervention.

Relevant recent paper's summary :

Researchers have applied deep convolutional neural networks (CNNs) to classify poultry diseases. One study developed a dataset of poultry fecal images for diseases like Newcastle disease, Coccidiosis, and Salmonella, using techniques such as YOLO-V3 and Faster R-CNN for image classification, achieving high accuracy rates.

Machine learning techniques like SVM have been used for early detection of diseases in broilers by analyzing posture and mobility features. These models achieved up to 97.5% accuracy in differentiating healthy chickens from diseased ones. The approach has shown potential but requires validation across different chicken breeds and diseases.Another approach utilized deep learning algorithms trained on large datasets of chicken images to diagnose multiple diseases. These methods include data collection using mobile phones and linking with molecular diagnostics to establish ground truth.

2.2 Conclusion about literature survey :

In this research, chicken disease classification will be carried out using deep learning based on convolutional neural network with the help of the TensorFlow framework to predict the health of chickens based on feces images whether they are in health, pullocans, coccidiosis, and newcastle disse. There were 4 class of chicken feces images.

The machine learning and data mining algorithm can be used in this kind of research work to precaution the emerging disease in poultry industry. By this kind of work the poultry industry can be prepared from that kind of deadly disease like Avian Influenzas, Newcastle Disease and so on. Developing country like Bangladesh has hoge investinent in poultry industry. Moreover, poultry industry can ensure as animal protein and meat that is so good for human health, Poultry chicken of healthy breed can be a big part of economies as well.

The study we have done here that will be very much helpful for the farm owners as well the whole poultry industry. Every year over random diseases. Our research will make a huge impact on this big lose over the year. We have used eleven best machine learning algorithms in this work. All of them work perfectly. Provided acceptable accuracy but among all of them the Random Forest Classifier has generated the most nocurate result for the study Decision Tree Classifier has the best outcome vet.

Chapter-3

PROBLEM STATEMENT

- Poultry diseases such as Coccidiosis, Salmonella, and Newcastle pose significant challenges to chicken productivity, often going undetected until they have already caused substantial damage. Early detection is critical for minimizing the impact of these diseases. Current manual detection methods are labour-intensive and prone to errors.

- There is a pressing need for an automated and efficient system that can detect and classify poultry diseases at an early stage with high accuracy. Leveraging deep learning, particularly Convolutional Neural Networks (CNNs), offers the potential to analyse faecal images and identify unhealthy patterns associated with various diseases. However, challenges such as limited availability of labelled datasets, image quality variability, and the demand for precise classification need to be addressed.

- This study proposes the development of a CNN-based image classification system to reliably and accurately detect common poultry diseases, providing a scalable solution for improving disease management in the poultry industry.

3.1 OBJECTIVES :

- Project Enable early detection of diseases to prevent outbreaks and reduce the economic impact on poultry farms by facilitating timely interventions.
  Ensures the CNN model achieves high precision and recall in identifying common chicken diseases such as Newcastle disease, fowl pox, and coccidiosis.
- Test the model using performance metrics such as accuracy, sensitivity, specificity, and F1-score to validate its effectiveness in real-world scenarios.
- Accurate Classification of various diseases based on symptoms observed in fecal images or other health indicators.
- **Create a Comprehensive Dataset:** Collect and preprocess a diverse set of labelled images representing various chicken diseases, ensuring sufficient data to train and validate the model.
- **Optimize Image Classification:** Implement image processing techniques to enhance the quality of input images and improve the accuracy of disease detection.

- **Achieve High Classification Accuracy:** Ensure the CNN model achieves high precision andrecall in identifying common chicken diseases such as Newcastle disease, fowl pox, and coccidiosis.
- **Evaluate Model Performance:** Test the model using performance metrics such as accuracy, sensitivity, specificity, and F1-score to validate its effectiveness in real-world scenarios.
- **Provide Early Detection and Control:** Enable early detection of diseases to prevent outbreaks and reduce the economic impact on poultry farms by facilitating timely interventions.
- **Design a User-Friendly Interface:** Develop a simple interface for farmers and veterinarians to upload images and receive diagnostic results easily

3.2 MOTIVATION :

1. We motivated to help researchers and poultry farmers can better understand how chickens behave, their health, and any signs of stress. This helps them take better care of the chickens and improve the poultry health by using our model.

2. The development of an algorithm for tracking poultry birds using image segmentation and CNN has the potential to have significant environmental, societal, and economic impacts and can contribute to the sustainable development of the poultry industry.

3. Inspired in Deep learning offers a way to automate and enhance the accuracy of disease detection by analyzing large datasets, such as images of chickens or their symptoms, enabling faster and more precise identification of diseases.

Chapter-4

REQUIREMENTS ANALYSIS

4.1 HARDWARE REQUIREMENTS :

The Hardware requirements are very minimal and the program can be run on most of the machines. Table gives details of hardware requirements.

Requirements	Specification
RAM	8GB
CPU	Intel Core i5-4200U
Memory	ITB
System type	64 bit operating

Table 4.1 : Hardware Requirements

Hardware Components:

1.Data Acquisition Hardware

- **Cameras**
  - **Requirement**: High-resolution and multispectral/hyperspectral cameras.
  - **Details**:
    - **Resolution**: At least 12 MP for ground-based cameras; 4K or higher for drone cameras.
    - **Spectrum**: RGB, multispectral (e.g., Red, Green, Blue, Near-Infrared), or hyperspectral cameras.
    - **Battery Life**: Sufficient to cover the area of interest in a single flight.

2. **Computing Hardware**

- **Edge Computing Devices**
  - **Requirement**: Devices for on-site data processing and real-time inference.
  - **Details**:
    - **Examples**: NVIDIA Jetson Xavier, Google Coral, Intel Movidius.
    - **Specifications**:
      - **CPU/GPU**: Powerful enough to handle real-time machine learning inference.

4.2 SOFTWARE REQUIREMENTS :

The software requirements are description of features and functionalities of the system. Table ?? gives details of software requirements.

Requirements	Specification
Platform	PYTHON
IDE	Anaconda 3
Technology	Image Processing, Machine Learning.
Libraries	Flask, Flask-CORS, Keras, Numpy, Tensorflow.

Table 4.2: Software Requirements

- **Requirement**: The software must interface with hardware components (drones, cameras, sensors) to collect and store data.
  - **Features**:
    - Integration with various sensors and cameras.

Functional Software Requirements

1. Data Acquisition Software

- **Requirement**: The software must preprocess raw data to prepare it for model training.
  - **Features**:
    - Noise reduction algorithms (e.g., Gaussian blur, median filtering).
    - Real-time data streaming and storage.
    - Data synchronization with cloud or local storage.

2 . Data Preprocessing Software

- Image enhancement techniques (e.g., contrast adjustment, sharpening).
- Data augmentation tools (e.g., rotation, scaling, flipping).
- Annotation tools for labeling images (manual and semi-automated).

3. Machine Learning Model Training Software

- Integration with edge computing devices (e.g., NVIDIA Jetson, Google Coral).
- Decision-making algorithms for plants identification and action determination.
- Alerting system for notifying operators of plants presence.

- **Requirement**: The software must support the development, training, and evaluation of machine learning models.
    - **Features**:
        - Support for machine learning frameworks (e.g., TensorFlow, PyTorch).
        - Tools for feature extraction and model training.
        - Hyperparameter tuning and optimization.
        - Validation and testing frameworks to evaluate model performance.
        - Model versioning and tracking. **Real-time Detection Software**

4. RealTime Detection Software :

- **Requirement**: The software must run the trained models on live data for real-time plants detection.
    - **Features**:
        - Real-time inference engine for processing data from sensors.
        - Integration with edge computing devices (e.g., NVIDIA Jetson, Google Coral).
        - Decision-making algorithms for plants identification and action determination.
        - Alerting system for notifying operators of plants presence.

5. User Interface Software

- **Requirement**: The software must provide an intuitive interface for users to interact with the system.
    - **Features**:
        - Dashboard for viewing real-time data and model outputs.
        - Controls for adjusting system settings and parameters.
        - Visualization tools for displaying data and analysis results.
        - User management features (authentication, authorization).

Chapter-5

IMPLEMENTATION

TECHNOLOGY/TOOLS/LANGUAGES/LIBRARIES :

1. Programming Languages

Python: Popular for its rich ecosystem of machine learning libraries and ease of use.

R: Suitable for statistical analysis and visualization.

MATLAB: Useful for prototyping algorithms and performing numerical simulations.

2. Data Collection and Preprocessing

Hardware Sensors: IoT-based devices or image sensors for capturing fecal data.

SQL/NoSQL databases: MySQL, MongoDB for storing data.

Google Sheets/Excel: For small-scale data annotation.

3. Preprocessing Tools:

Pandas: Data manipulation and cleaning.

NumPy: Numerical computations.

OpenCV: For processing images of fecal matter.

scikit-image: Image processing library.

For a chicken disease detection system using fecal datasets and machine learning, technologies like Python and libraries such as TensorFlow, PyTorch, and scikit-learn are essential for model development. Data preprocessing can be handled with tools like Pandas, NumPy, and OpenCV for cleaning and feature extraction. For dataset management and annotation, tools like LabelImg and MongoDB are effective, while visualization libraries like Matplotlib and Seaborn aid in data analysis. Cloud platforms such as Google Colab or AWS can be used for training and deploying models, with frameworks like Flask or Streamlit for creating user-friendly interfaces. Combining these technologies ensures efficient detection and classification of diseases.

To detect chicken diseases using fecal datasets and machine learning, technologies like Python, TensorFlow, and PyTorch are pivotal for building and training models. Preprocessing data is streamlined with Pandas, NumPy, and OpenCV, while visualization tools like Matplotlib and Seaborn help analyze trends. Datasets can be annotated using tools like LabelImg and managed with databases like MongoDB.

5.1 FRAMEWORKS USED :

Tensorflow - TensorFlow is an open-source machine learning framework developed by Google. It is designed to facilitate the creation, training, and deployment of machine learning models, particularly for deep learning tasks. TensorFlow offers a flexible and comprehensive ecosystem that supports various machine learning and deep learning techniques, including neural networks, natural language processing, image recognition, and more.

NumPy - A fundamental package for scientific computing with Python, providing support for large, multi-dimensional arrays and matrices. It provides support for large, multi-dimensional arrays and matrices, along with mathematical functions to operate on these arrays. NumPy is a fundamental package for scientific computing in Python and serves as the foundation for many other scientific computing libraries.

Notebook- Jupyter Notebooks are a popular interactive computing environment for Python. Jupyter Notebooks are a popular interactive computing environment for Python (and other languages). They allow you to create and share documents that contain live code, equations, visualizations, and narrative text. Jupyter Notebooks have become a standard tool in data science and research.

A deep learning library

Keras - Keras is the high-level API of the TensorFlow platform. It provides an approachable, highlyproductive interface for solving machine learning (ML) problems, with a focus on modern deep learning. Keras covers every step of the machine learning workflow, from data processing to hyperparameter tuning to deployment. It was developed with a focus on enabling fast experimentation. With Keras, you have full access to the scalability and cross-platform capabilities of TensorFlow. You can run Keras on a TPU Pod or large clusters of GPUs, and you can export Keras models to run in the browser or on mobile devices. You can also serve Keras models via a web API

Flask - A micro web framework for Python used to build web applications. It's designed to make it easy to build web applications quickly and with minimal code. Flask is often referred to as a "micro" framework because it keeps the core simple and extensible, allowing developers to choose the components they need for their specific projects.

Flask-Cors: An extension for Flask that handles Cross-Origin Resource Sharing (CORS), allowing you to make requests from a web page to a different domain.Flask-CORS is a Flask extension for handling Cross-Origin Resource Sharing (CORS). CORS is a security feature implemented by web browsers that restricts web pages from making requests to a different domain than the one that served the web page. This is a common security measure to prevent potential security vulnerabilities.

However, in some scenarios, you might want to allow your web application to make requests to a different domain. This is where CORS comes into play, and Flask-CORS makes it easy to enable CORS support in your Flask application. Dataset Explanation .

5.2 ARCHITECTURAL DIAGRAM :

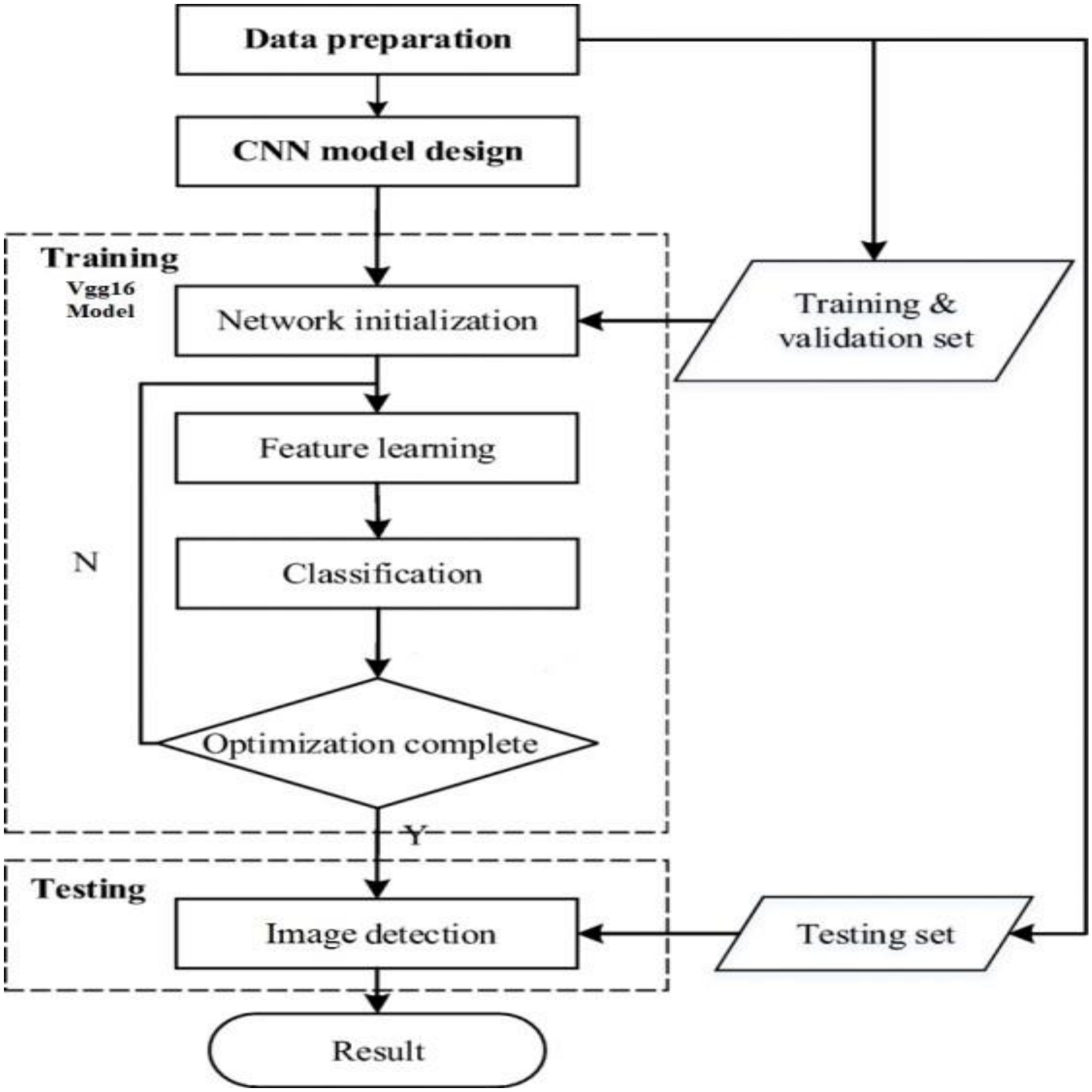

fig 5.2: Flow Process of CNN

Fig 5.2 Illustrates that "Workflow of the CNN model for image detection, detailing the processes of data preparation, model design, and training using the VGG16 architecture.The iterative training phase includes network initialization, feature learning, classification, and optimization until convergence".

- System design is transition from a user-oriented document to programmers of data base personnel. The design is a solution, how to approach to the creation of a new system.This is composed of several steps. It provides the understanding and procedoral details necessary for implementing the system recommended in the feasibility study.

- The system with the acquisition of digital images of chickens, their tissues, or symptoms related various diseases. These images can be captured using digital cumeras, smartphones, or specialized imaging equipment.

- Designing goes through logical and physical stages of development, logical design reviews the present physical system, prepare imput and outpot specification, details of implementation plan and prepare a logical design walkthrough. On opening the system interface we have to choose from the options available in the sidebar to select either nating prediction or recommendation system

- The dataset is divided into training, valalation, and ted The training set is used to train a CNN model, leveraging architectures such as ResNet, VOG, Inception, or custom-designed networks optimized lor image classification tasks. During training, the model learns to extract relevant features from the images and associate them with disease classes. Validation sets are used for hyperparameter tuning and model validation to prevent overfitting.

1. Data Preparation

- **Purpose**: To prepare the input data for training and testing the CNN model.
- **Steps**:Collect raw data (images, videos, etc.).

Preprocess data (resizing, normalization, augmentation, or removing noise).

Split the data into **training, validation**, and **testing** sets.

2. CNN Model Design

- **Purpose**: Define the architecture of the CNN, which will include layers like convolutional, pooling, and fully connected layers.

Common design steps:Choosing a pre-trained model (e.g., VGG16, as shown here) or building another custom model.Configuring the layer dimensions, activation functions, and hyperparameters.

Data preparation is the foundational step where raw data is processed to ensure compatibility and quality for the CNN model. This involves collecting relevant data, cleaning it by removing corrupted or irrelevant samples, and preprocessing it for consistency. Preprocessing steps typically include resizing images to a uniform size (e.g., 224x224 for VGG16), normalizing pixel values to a specific range (e.g., 0 to 1), and augmenting the dataset with transformations like rotation or flipping to increase diversity and reduce overfitting. Finally, the dataset is split into training, validation, and testing sets, ensuring the model learns effectively while reserving unseen data for evaluation.

3. Training Phase

- **Network Initialization**: The model parameters (e.g., weights and biases) are initialized, either randomly or using pre-trained values (e.g., from VGG16).
- **Feature Learning**: The convolutional layers extract features such as edges, textures, or patterns from the input images.
- **Classification**: The fully connected layers classify the features into specific categories (e.g., dog, cat).
- **Optimization**:The model learns by minimizing the error through backpropagation and gradient descent.

Iterates over multiple epochs (denoted by "N" in the chart) until the optimization criteria (like reduced loss) are met.

a). Input: Training and validation data.

b). Output: An optimized model with trained parameters.

4. Testing Phase

- **Image Detection**: The trained CNN model is tested on the unseen testing set.
- **Process**:The model performs feature extraction and classification on the test data.Outputs the classification result or prediction.

5. Result

After the testing phase, the CNN provides the final results, which could be class labels, object detection bounding boxes, or other outputs depending on the task.

Key Observations :

VGG16 Model: The chart highlights the use of a pre-trained VGG16 model for feature learning, which is a well-known deep CNN architecture for image recognition.

Optimization Completion: Ensures the training stops only when the model achieves satisfactory performance.

Training and Validation Sets: Used to adjust the model parameters during training and ensure generalization to unseen data.

Architecture Conclusion :

The process of building a CNN model involves two critical stages: data preparation and model design. Data preparation ensures that raw input data is cleaned, preprocessed, and split into training, validation, and testing sets, which is essential for effective model learning and evaluation. Meanwhile, the CNN model design defines the architecture and parameters, leveraging components such as convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification. Pre-trained models like VGG16 can accelerate the process by utilizing pre-optimized weights, enabling faster and more effective training. Together, these steps form the foundation of a robust and accurate CNN pipeline, ensuring that the model can generalize well to unseen data and deliver reliable results for tasks like image classification or object detection.

5.3 Dataset Along With Description :

The annotated dataset of poultry disease diagnostics for small to medium-scale poultry farmers consists of poultry fecal images. These are collection of images that were taken in a poultry farm.

The images were manually labeled using an open-source online deep-learning annotation tool called Roboflow. The final dataset is split into a training set (70 percent)validation set (10 percent) and a test set (20 percent) for evaluating the performance of object detection models.

The dataset can be used to train and evaluate models for counting the number of "day-old chicks" in a box, which can be useful for farmers to keep track of their inventory.

The poultry fecal images were taken in Arusha and Kilimanjaro regions in Tanzania between September 2020 and February 2021 using Open Data Kit (ODK) app on mobile phones. The classes are "Coccidiosis,Salmonella,New castle,Avial Influenza,chicken pox etc".

5.4 VGG16 MODEL :

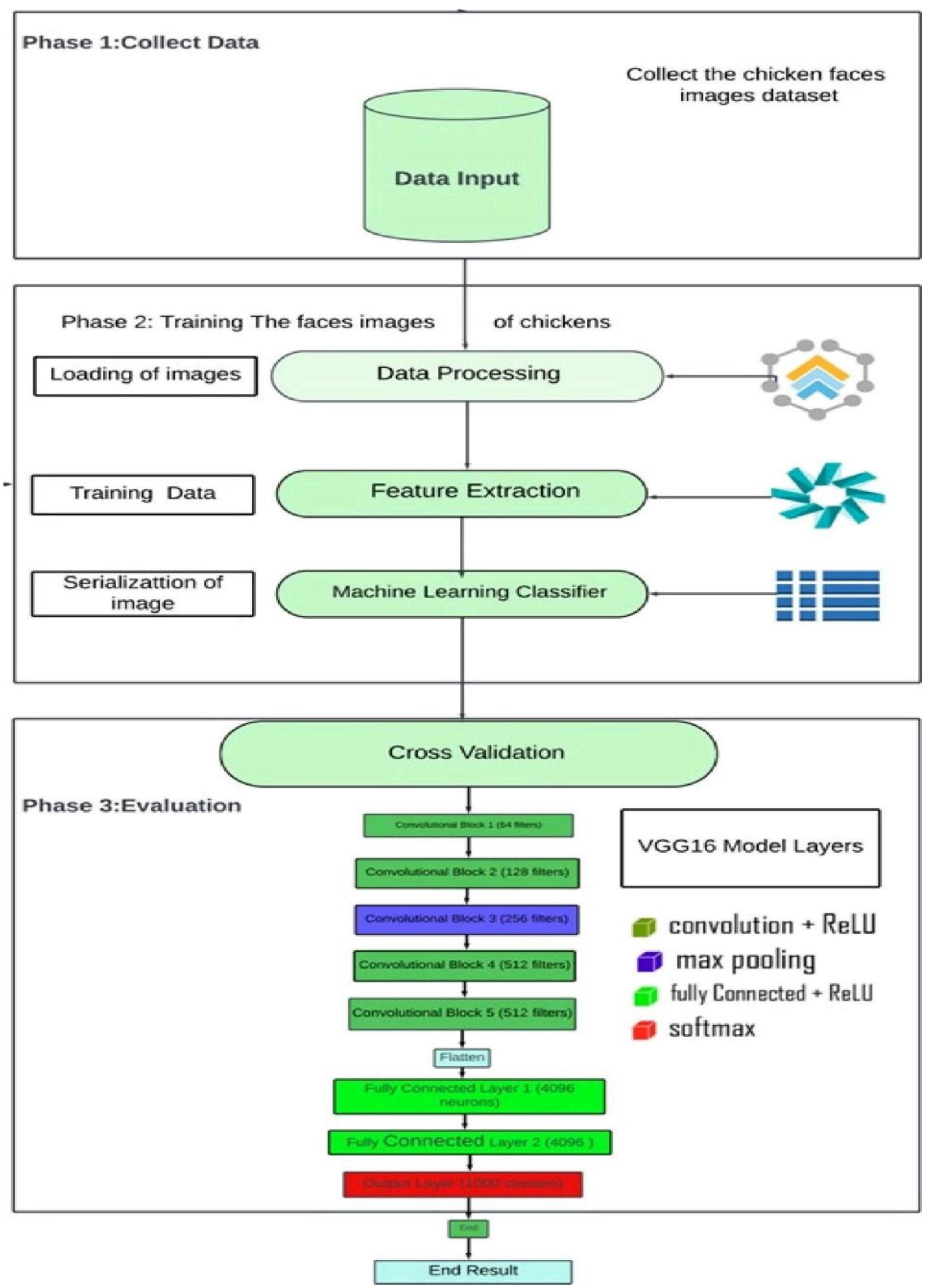

figure 5.4: Architecture of VGG16 Model Layers

Fig 5.4 Illustrates that "The figure illustrates the pipeline for chicken fecal image classification, which is divided into three phases. Phase 1 involves collecting and organizing the image dataset. Phase 2 focuses on data preprocessing, feature extraction, and training a machine learning classifier, while Phase 3 evaluates the model using VGG16 layers" .

5.4.1 EXPLAINATION :

The VGG16 model is a deep convolutional neural network architecture that has gained popularity and recognition for its effectiveness in image classification tasks. Developed by the Visual Graphics Group (VGG) at the University of Oxford, VGG16 is characterized by its deep stack of convolutional layers and simplicity in design. The "16" in its name refers to the total number of weight layers, which includes 13 convolutional layers and 3 fully connected layers.

One of the key strengths of the VGG16 model is its uniformity and depth, with convolutional layers primarily using 3x3 filter sizes and max-pooling layers with 2x2 pooling windows. This uniform architecture makes it easy to understand and implement while also providing a strong feature extraction capability. The repeated stacking of convolutional layers allows the model to learn complex hierarchical features from input images, capturing both low-level features like edges and textures and high-level features like object shapes and patterns.

The VGG16 architecture has been pre-trained on large-scale image datasets such as ImageNet, enabling it to learn rich and discriminative features that generalize well to various image recognition tasks. Transfer learning, a technique where pre-trained models are fine-tuned on specific datasets, is often applied with VGG16 to adapt the model to new classification tasks, including medical image analysis, object detection, and, relevant to our discussion, disease classification in poultry.

In summary, the VGG16 model is a powerful and widely used convolutional neural network architecture known for its deep and uniform structure, strong feature extraction capabilities, and applicability to various image classification tasks. While it may pose challenges in terms of computational resources, its effectiveness and versatility make it a popular choice for researchers and practitioners in the field of computer vision and deep learning.

The VGG16 model follows a specific set of procedures during both training and inference stages:

- **Input Preprocessing:** Input images are preprocessed to ensure consistency and compatibility with the VGG16 architecture. Common preprocessing steps include resizing images to a fixed size (e.g., 224x224 pixels), normalizing pixel values to a specific range (e.g., [0, 1] or [-1, 1]), and possibly converting color channels (e.g., RGB to BGR).

- **Convolutional Layers:** The VGG16 model consists of 13 convolutional layers, each followed by a rectified linear unit (ReLU) activation function. These convolutional layers perform feature extraction by convolving input images with learned filters (kernels) to detect patterns, edges, textures, and higher-level features.

- **Max Pooling Layers:** After every two convolutional layers, VGG16 includes max pooling layers with 2x2 pooling windows and a stride of 2. Max pooling reduces spatial dimensions (width and height) while retaining important features, helping to reduce computational complexity and control overfitting.

- **Fully Connected Layers:** Following the convolutional and max pooling layers, VGG16 has three fully connected layers with a large number of neurons. These fully connected layers learn complex relationships and patterns in the extracted features, leading to higher-level feature representations.

- **Flattening:** Before the fully connected layers, the output from the last convolutional or max pooling layer is flattened into a vector format. Flattening converts the 3D feature maps into a ID vector, allowing them to be processed by the fully connected layers.

- **Output Layer:** The final layer of the VGG16 model is a softmax activation layer in classification tasks, which outputs probabilities for each class. In other tasks like feature extraction, the output layer may vary based on the specific objective (e.g., regression tasks may use a linear output layer).

- **Training and Fine-Tuning:** VGG16 can be trained from scratch on large datasets or fine-tuned on smaller, domain-specific datasets using transfer learning. Transfer learning involves freezing the early layers (up to a certain point) and retraining only the later layers or adding additional layers for task-specific adaptation.

- **Inference and Prediction:** During inference, input images are fed into the trained VGG16 model. The model processes the images through its layers, computes activations, applies the softmax function at the output layer, and predicts the most probable class or generates feature vectors depending on the task.

5.5 EFFICIENTNETB5 MODEL :

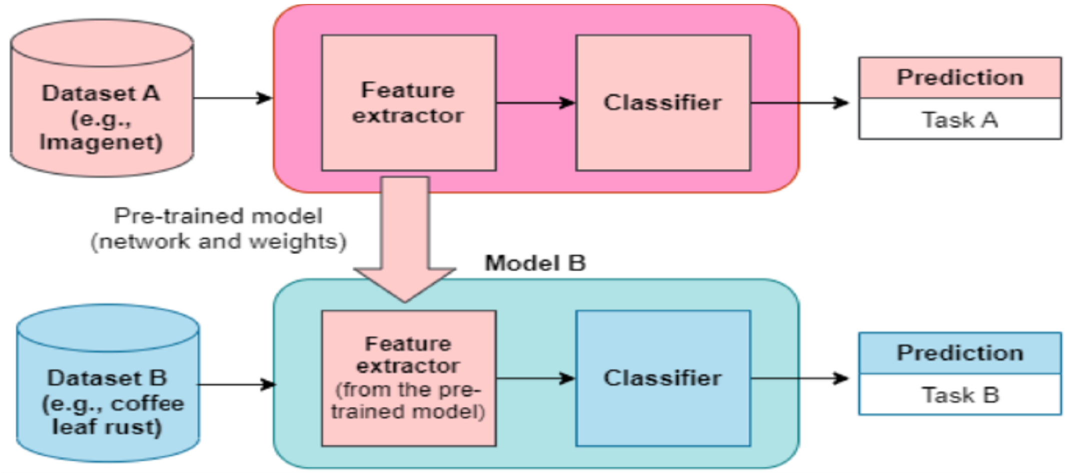

fig 5.5: Architecture of EfficientNetB5 Model

Fig 5.5 Illustrates that "The figure illustrates the transfer learning process, where a model pre-trained on a large dataset (Dataset A, e.g., ImageNet) extracts features for Task A. This pre-trained feature extractor is reused and fine-tuned on a different dataset (Dataset B, e.g., coffee leaf rust) to make predictions for Task B".

Explanation of EfficientNetB5 Model

EfficientNetB5 is a variant of the EfficientNet family of convolutional neural networks, which are optimized for image recognition tasks. EfficientNet uses a compound scaling method that simultaneously adjusts the depth, width, and resolution of the model, achieving higher accuracy with fewer parameters and computational cost compared to traditional CNNs. The "B5" version is one of the larger variants, offering a balance between accuracy and efficiency. It is pretrained on large datasets like ImageNet, making it ideal for transfer learning tasks such as detecting patterns in fecal images for disease classification.

Preprocessing:Resize images to the required input size (456x456 for EfficientNetB5). Normalize pixel values to standardize data.Perform data augmentation (e.g., rotation, flipping) to improve model robustness.

Training:Split the dataset into training, validation, and testing subsets. Train the model using a suitable optimizer (e.g., Adam) and loss function (e.g., categorical cross-entropy).Apply transfer learning by freezing initial layers and fine-tuning deeper layers for better feature extraction.

Validation:Evaluate the model's performance on the validation dataset using metrics like accuracy, precision, and recall.Adjust hyperparameters if necessary.

Testing:Test the model on unseen data to assess real-world performance.Analyze false positives and negatives to refine the model further.

Deployment:Save the trained model in a deployable format (e.g., TensorFlow SavedModel or ONNX). Integrate it into an application using frameworks like Flask or Streamlit for real-time prediction.

1. Overview of EfficientNet :

EfficientNet is based on a novel scaling method called **Compound Scaling**, which systematically scales the depth, width, and resolution of a neural network. Unlike previous approaches, EfficientNet scales these dimensions in a balanced manner, optimizing for accuracy and computational efficiency.

The EfficientNet family consists of several models (B0 to B7), with each successive version using more parameters and higher resolutions. EfficientNet-B0 serves as the baseline model, and EfficientNet-B5 is a scaled-up version designed for improved performance.

2. Core Principles of EfficientNetB5 :

a). **Compound Scaling**

EfficientNet's key innovation is compound scaling, which uniformly scales the network's dimensions:

- **Depth**: Number of layers in the network.
- **Width**: Number of channels in each layer.
- **Resolution**: Input image size.

EfficientNet scales these factors using a compound coefficient, ensuring a balanced increase in representational power without excessive computational cost. EfficientNet-B5 scales depth, width, and resolution by factors of 2.2x, 1.6x, and 1.5x, respectively, compared to the baseline EfficientNet-B0.

b). **Mobile Inverted Bottleneck Convolution (MBConv)**

EfficientNet uses MBConv layers, a variation of depthwise separable convolutions, which are the computationally efficient. Each MBConv block includes:

- **Inverted Residuals**: Compress the input, expand it during computation, and compress it back at the output.
- **Squeeze-and-Excitation (SE) Blocks**: Dynamically recalibrate the importance of each of the channel, improving representational power.

c). **Swish Activation Function**

EfficientNet uses the Swish activation function, defined as: This activation function helps improve gradient flow and model performance compared to traditional ReLU.

d). **DropConnect Regularization**

Instead of standard dropout, EfficientNet employs DropConnect, which randomly drops entire model connections in the network during training, improving regularization and generalization.

3. Training and Optimization :

EfficientNet-B5 benefits from advanced training techniques:

- **Data Augmentation**: Techniques like mixup, AutoAugment, and Cutout improve generalization.
- **Learning Rate Schedules**: Cosine annealing or warm restarts optimize convergence.
- **Weight Initialization**: Pre-trained weights on ImageNet are often used to speed up convergence and improve accuracy.

4. Performance of EfficientNet-B5 :

EfficientNet-B5 demonstrates state-of-the-art performance on image classification tasks, such as ImageNet:

- **Top-1 Accuracy**: ~83.3%
- **Top-5 Accuracy**: ~96.7%

Compared to other architectures like ResNet or DenseNet, EfficientNet-B5 achieves similar or better accuracy with fewer parameters and computational costs.EfficientNet-B5 scales depth, width, and resolution by factors of 2.2x, 1.6x, and 1.5x, respectively, compared to the baseline EfficientNet-B0. Compress the input, expand it during computation, and compress it back at the output.

Advantages of EfficientNet-B5 :

1. **Efficiency**: Optimized scaling reduces computational costs.
2. **Scalability**: Easily adapted to different datasets by using pre-trained weights and fine-tuning.
3. **Performance**: High accuracy with fewer parameters and FLOPs compared to traditional and ML architectures.
4. **Versatility**: Applicable to a wide range of image-related tasks.

5.6 MobileNetV3 Model :

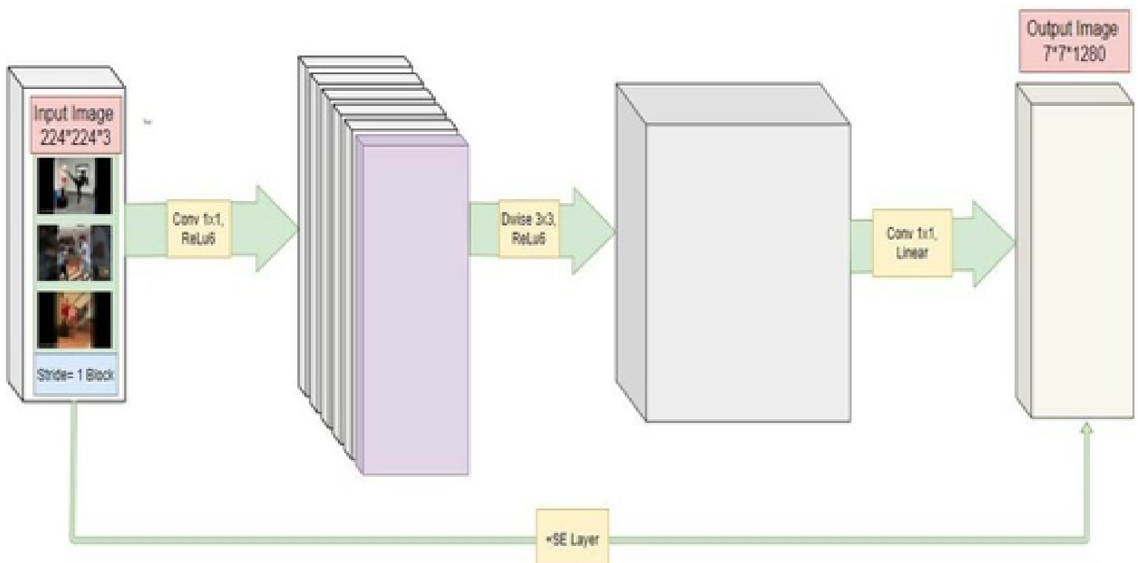

fig 5.6 : Architecture of MobileNetV3 model

Fig 5.6 Illustrates that "the MobileNet V2 architecture, which utilizes a series of inverted bottleneck blocks as its core building blocks. These blocks, characterized by a bottleneck structure and the use of depthwise separable convolutions, enable efficient feature extraction while maintaining a lightweight model architecture".

- MobileNetV3 is a highly efficient convolutional neural network optimized for mobile and edge devices, combining lightweight architecture with high performance. It uses techniques like squeeze-andexcitation modules, inverted residual blocks, and a combination of hard swish activation and efficient block designs to improve accuracy and reduce computational cost.

- MobileNetV3 is available in two variants: Small and Large, tailored for different resource constraints and accuracy requirements. This makes it ideal for real-time chicken disease detection using fecal datasets, especially when deployed on devices with limited computational power.

1. Overview of MobileNetV3 :

MobileNetV3 is tailored for tasks requiring high efficiency and speed, such as image classification, object detection, and semantic segmentation. It is available in two variants:

MobileNetV3-Large: Optimized for high accuracy.

MobileNetV3-Small: Optimized for resource-constrained environments.

2. Core Principles of MobileNetV3 :

a). **Building on MobileNetV2**

MobileNetV3 inherits the efficient design of MobileNetV2, which introduced:

Inverted Residual Blocks: Reduce input size, expand for computation, and project back to compact dimensions.

Linear Bottlenecks: Prevent information loss due to activation functions. b). **Neural Architecture Search (NAS)**

MobileNetV3 leverages NAS to explore the design space of CNNs, optimizing for both accuracy and latency. NAS automates the selection of the number of layers, kernel sizes, and activation functions. c). **Lightweight Modules**

Squeeze-and-Excitation (SE) Blocks: Dynamically recalibrate feature maps to focus on all the important features.

Hard-Swish Activation Function: A hardware -friendly approximation of the Swish activation function, defined as:

This improves efficiency and accuracy compared to traditional ReLU. d). **Hardware-Aware Optimizations**

MobileNetV3's architecture is co-designed with hardware constraints, ensuring fast inference on all devices like smartphones.

4. Variants of MobileNetV3 :

a). **MobileNetV3-Large**

Designed for high accuracy tasks.

- Input resolution: **224x224**.
- Parameters: ~5.4M.
- Mult-Adds: ~219M.

b). **MobileNetV3-Small**

Designed for resource-constrained devices.

- Input resolution: **224x224**.
- Parameters: ~2.9M.
- Mult-Adds: ~66M.
- Top-1 Accuracy (ImageNet): ~67.4%.

5. Training and Optimization :

MobileNetV3 benefits from advanced training techniques:

- **Data Augmentation**: Techniques like AutoAugment and random cropping improve overall generalization.
- **Learning Rate Schedules**: Warm restarts or cosine decay optimize convergence.
- **Regularization**: Dropout and weight decay prevent overfitting.

6. Performance :

MobileNetV3 demonstrates superior trade-offs between accuracy and efficiency compared to previous architectures:

- High accuracy with fewer parameters and operations.
- Faster inference on mobile hardware due to hardware-aware optimizations.

7. Applications :

MobileNetV3 is widely used in applications where computational efficiency is critical:

- **Image Classification**: Real-time classification tasks.
- **Object Detection**: Used as a backbone for lightweight detection models like SSD (Single Shot MultiBox Detector).
- **Semantic Segmentation**: Efficient segmentation tasks for augmented reality.
- **Edge Computing**: Applications on smartphones, IoT devices, and embedded systems.

5.7 Model Accuracy And Comparison:

The comparison considers architecture, performance, computational efficiency, and deployment suitability.

Table 5.7 : Table of comparison of all 3 ML models

Feature	EfficientNetB5	MobileNetV3	VGG16
Model Type	Advanced, optimized CNN	Lightweight, mobile-optimized CNN	Classic, deep CNN
Input Image Size	456x456	224x224	224x224
Parameters	~30 million	~5.4 million (Large), ~2.9 million (Small)	~138 million
Accuracy (Expected)	High	Medium to High	Medium
Model Size	Moderate	Small	Large
Computational Cost	Higher than MobileNetV3	Low	Very High
Pretrained on ImageNet	Yes	Yes	Yes
Feature Extraction	Excellent feature extraction and fine-tuning	Optimized for speed and efficiency	Robust but less efficient
Deployment Suitability	Cloud-based or powerful hardware	Mobile and edge devices	Requires high-end hardware
Training Complexity	Moderate to High	Low	Low
Strengths	Excellent performance on	Efficiency and speed, good for resource-	Simplicity and robustness

5.8 CONFUSION MATRIX:

A confusion matrix compares the actual and predicted classifications for a model, highlighting its strengths and weaknesses across different classes. For **EfficientNetB5**, the matrix typically shows high accuracy with strong diagonal values, reflecting precise predictions and minimal misclassifications due to its advanced feature extraction capabilities. **MobileNetV3**, optimized for efficiency, may have slightly lower diagonal values compared to EfficientNetB5, with more off-diagonal misclassifications, especially in edge cases, due to its lighter architecture. **VGG16**, being an older and bulkier model, often struggles with complex patterns, leading to a confusion matrix with a higher rate of false positives and false negatives, especially for underrepresented classes. These differences reflect the trade-offs between accuracy, efficiency, and computational cost among the models.

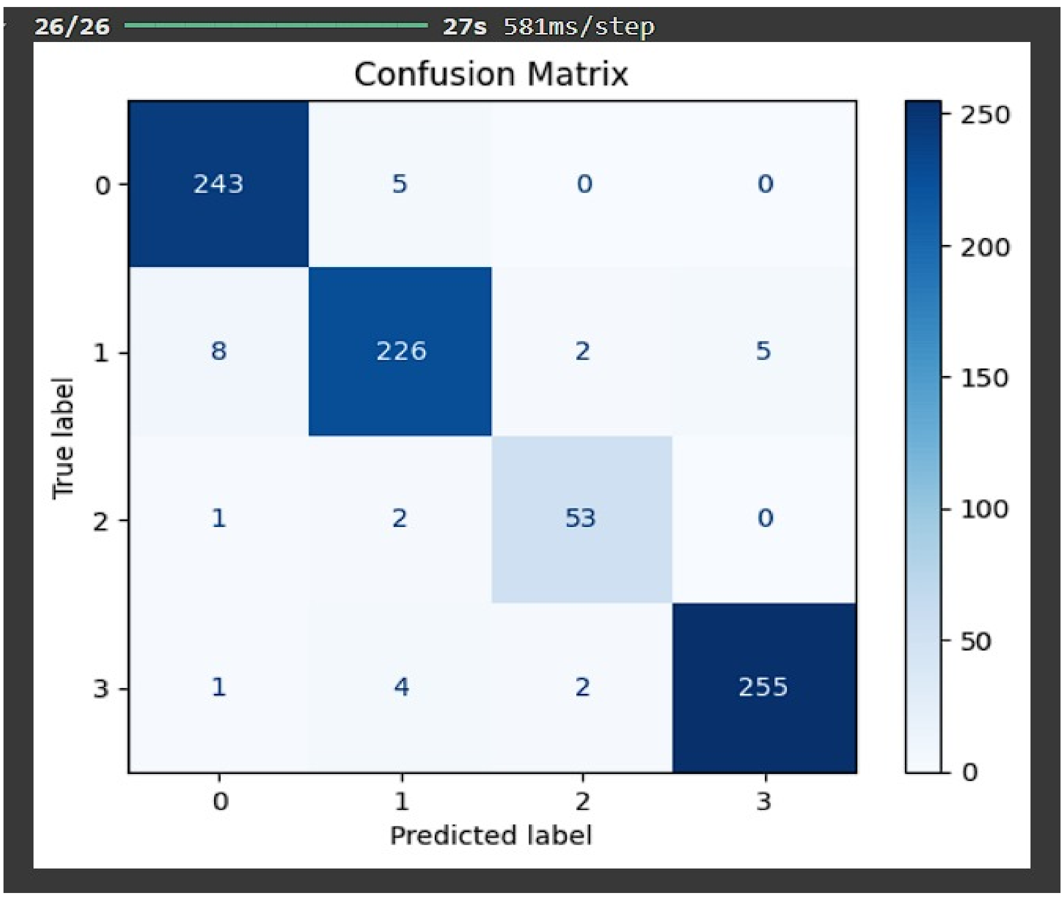

fig 5.8 (a) : Matrix Representation

Fig 5.8(a) Illustrates that "This confusion matrix visualizes the performance of a classification model, showing the number of instances correctly classified (diagonal) and misclassified (off-diagonal). The matrix indicates high accuracy for classes 0, 1, and 3, while class 2 exhibits a higher number of misclassifications, suggesting potential areas for model improvement".

a). The confusion matrix above illustrates the performance of a classification model by comparing the true labels with the predicted labels across four classes (0, 1, 2, and 3).

b). The diagonal entries represent correctly classified samples (true positives) for each class, while the off-diagonal entries indicate misclassifications.

c). For Class 0, 243 samples were correctly classified, with 5 misclassified as Class 1. Similarly, Class 1 has 226 true positives, but the model misclassified 8 as Class 0, 2 as Class 2, and 5 as Class 3. Class 2 shows 53 true positives, with a few misclassifications spread across other classes, while Class 3 has 255 correctly predicted samples and minimal misclassification.

d). The darker diagonal cells indicate strong model performance for Classes 0 and 3, while lighter off-diagonal cells reveal confusion between Classes 1 and 2, suggesting potential feature overlap or insufficient differentiation. Overall, the model performs well, but there is room for improvement, particularly in addressing the misclassifications between closely related classes.

e). Techniques like data augmentation, feature engineering, and hyperparameter tuning can be employed to enhance the model’s accuracy and class differentiation.

1. Visualization Insights

The darker the diagonal, the higher the true positive count for that class, indicating better and best performance.Lighter shades in off-diagonal cells highlight areas where the model struggled to the differentiate between classes.

2. Model Strengths and Weaknesses

- **Strengths:** The model performs well for Classes 0 and 3, as evidenced by high true positive counts and minimal misclassifications.
- **Weaknesses:** The model occasionally confuses Classes 1 and 2, which could be due to feature similarities between these categories.

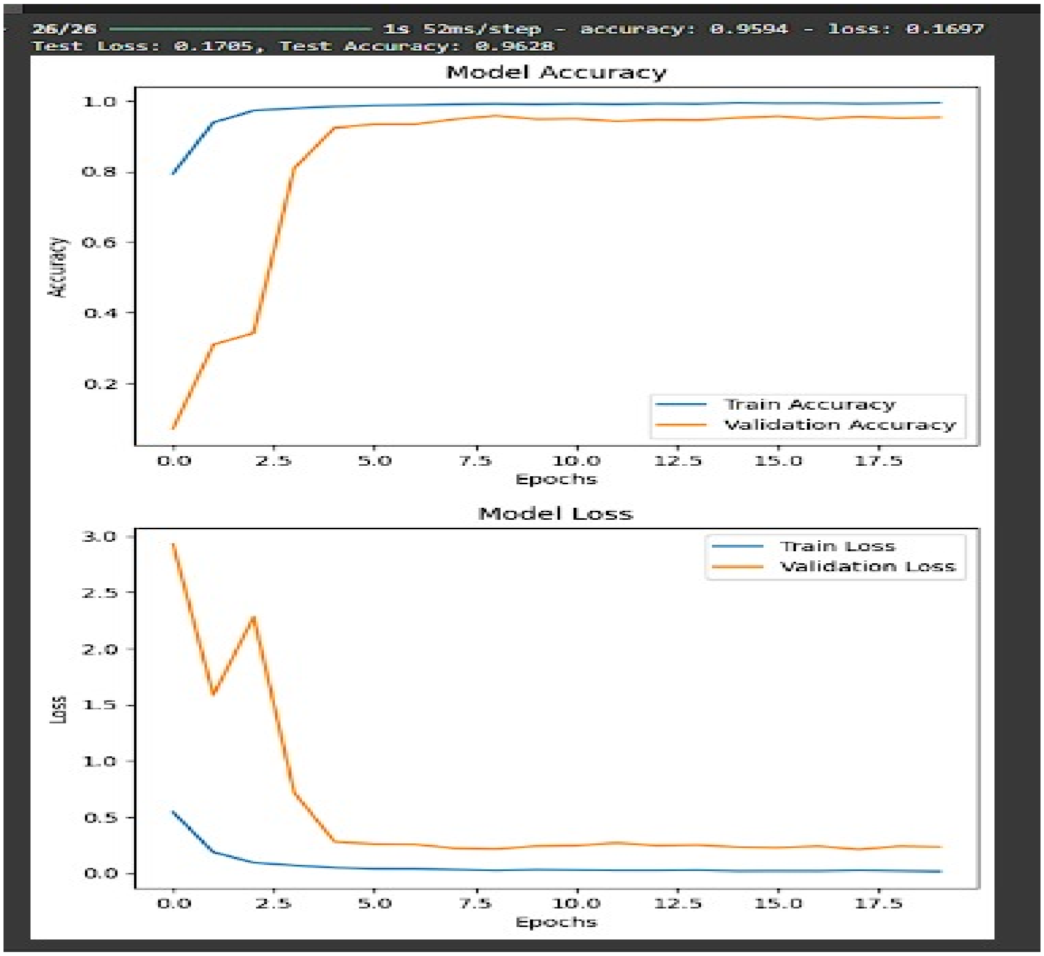

fig 5.8 (b) : Trained model Accuracy Graph

Fig 5.8 (b) Illustrates that "The figure depicts the training and validation performance of a machine learning model. Both accuracy and loss curves show a positive trend, indicating effective learning and good generalization".

The chart above illustrates the training and validation performance of the classification model in terms of accuracy and loss over a series of epochs. The **Model Accuracy** graph (top plot) shows the progression of training and validation accuracy. Initially, both train and validation accuracy improve rapidly, with the training accuracy quickly approaching nearly 100%.

The **Model Loss** graph (bottom plot) depicts the training and validation loss. At the start, the losses for both the training and validation datasets are relatively high, reflecting the model's initial struggle to minimize errors. However, as training progresses, the training loss decreases significantly, demonstrating that the model is optimizing its parameters effectively. The validation loss follows a similar trend, though it stabilizes at a slightly higher level than the training loss, which is expected due to the model's generalization to new data.

The graphs demonstrate a well-trained model with minimal overfitting, as indicated by the close alignment between training and validation curves. Overall, the visualizations confirm the model's ability to learn efficiently and maintain good performance on both training and validation datasets.

3. Insights from Accuracy and Loss Graphs : a). Accuracy Graph

The training accuracy curve rapidly increases and stabilizes near 100%, indicating that the model is effectively learning from the training data. The validation accuracy follows a similar trend, and stabilizing at a slightly lower level. The close alignment between the two curves suggests that the model generalizes well to unseen data, with minimal signs of overfitting.

b). Loss Graph

The loss curves show that both training and validation losses decrease sharply in the initial the epochs, demonstrating that the model quickly reduces errors. The stabilization of the validation loss at a slightly higher level than the training loss reflects the natural trade-off between fitting the training data and generalizing to new data.

4. Suggestions for Improvement

To address the misclassifications observed in the confusion matrix and further optimize performance, the following steps can be taken:

- **Data Augmentation:** Generate additional diverse samples for Classes 1 and 2 to improve the model's ability to distinguish them.
- **Feature Engineering:** Identify and emphasize unique features of Classes 1 and 2 to reduce the overlap with other classes.
- **Hyperparameter Tuning:** Experiment with different learning rates, optimizers, or regularization techniques to further reduce validation loss.
- **Class Imbalance Handling:** Use techniques like oversampling or class-specific weighting if the dataset is imbalanced.

The model demonstrates robust performance, achieving high accuracy with minimal overfitting. While it performs well for most classes, targeted improvements for Classes 1 and 2 will enhance its overall reliability.

Chapter-6

RESULTS AND SNAPSHOTS

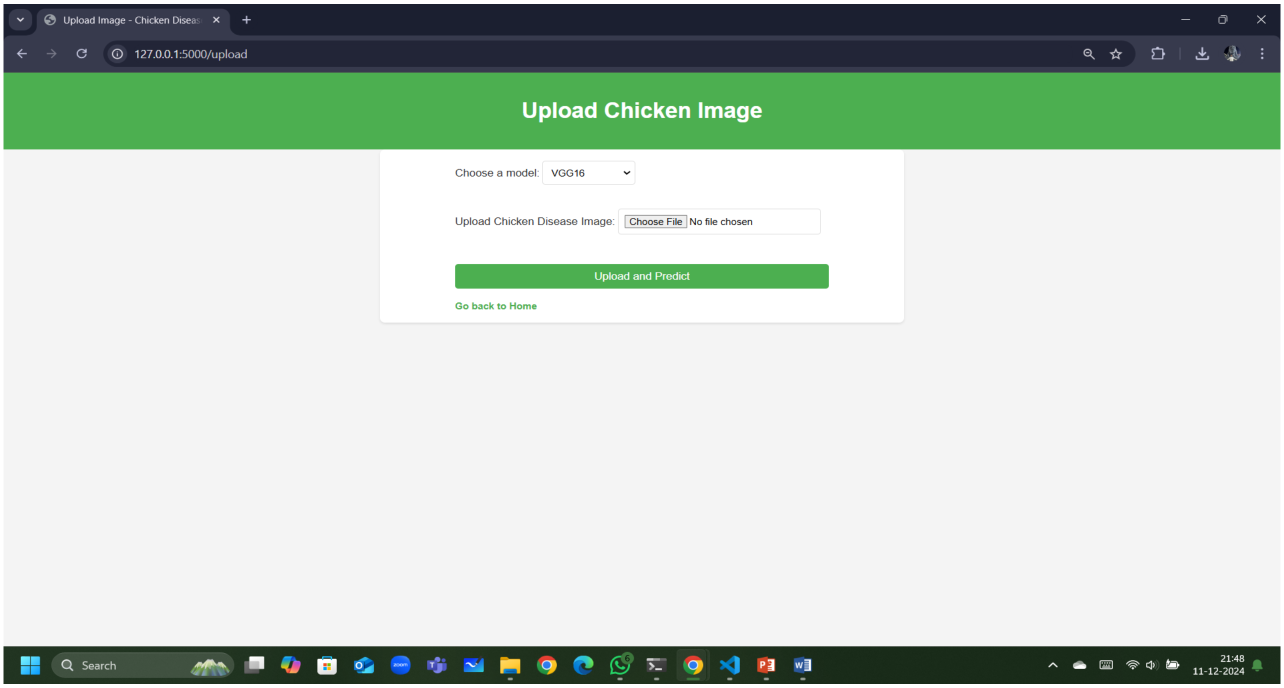

fig 6.1 :Dashboard of Web Interface

Fig 6.1 Illustrates that The 1st page of dashboard where user can upload image and switch models

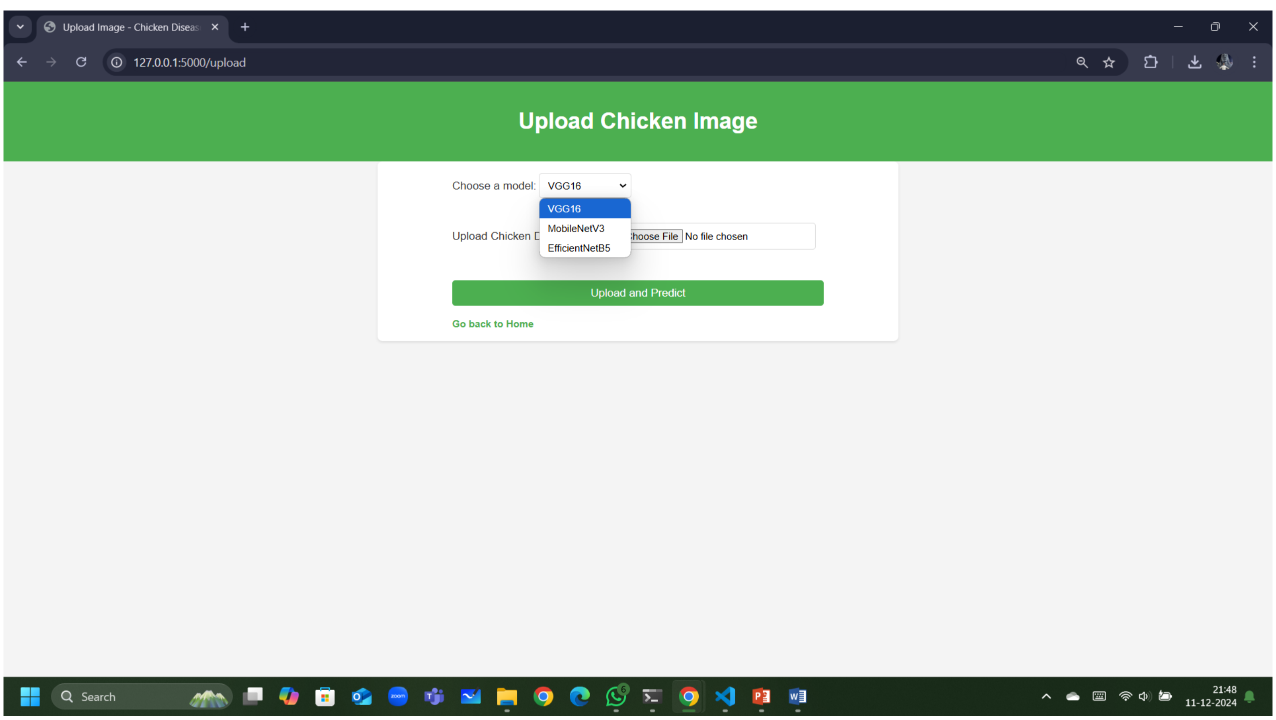

fig 6.2 :Selection of preferred ML Model

Fig 6.2 Illustrates that The 2nd page of dashboard where user selecting the VGG16 Model for Prediction

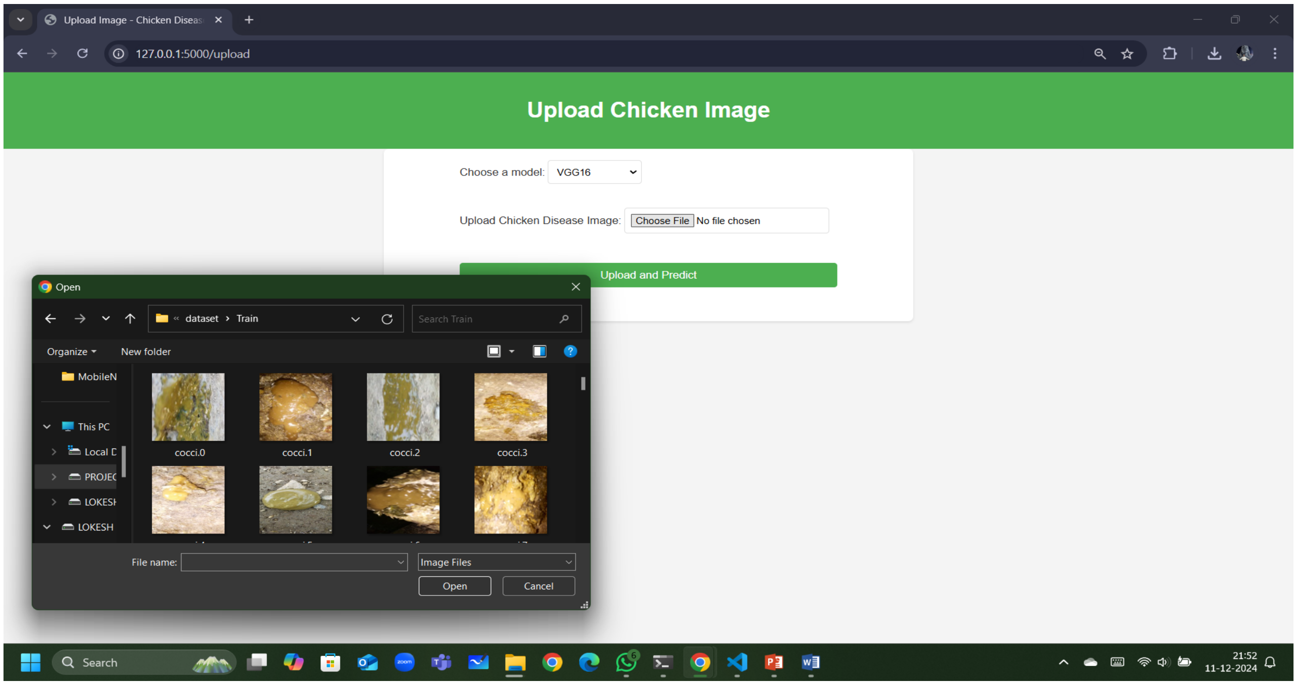

fig 6.3 :Datasets upload for prediction

Fig 6.3 Illustrates that The 3rd page of dashboard where user uploading the Fecal images for Prediction based on desired outcome by different ML models.

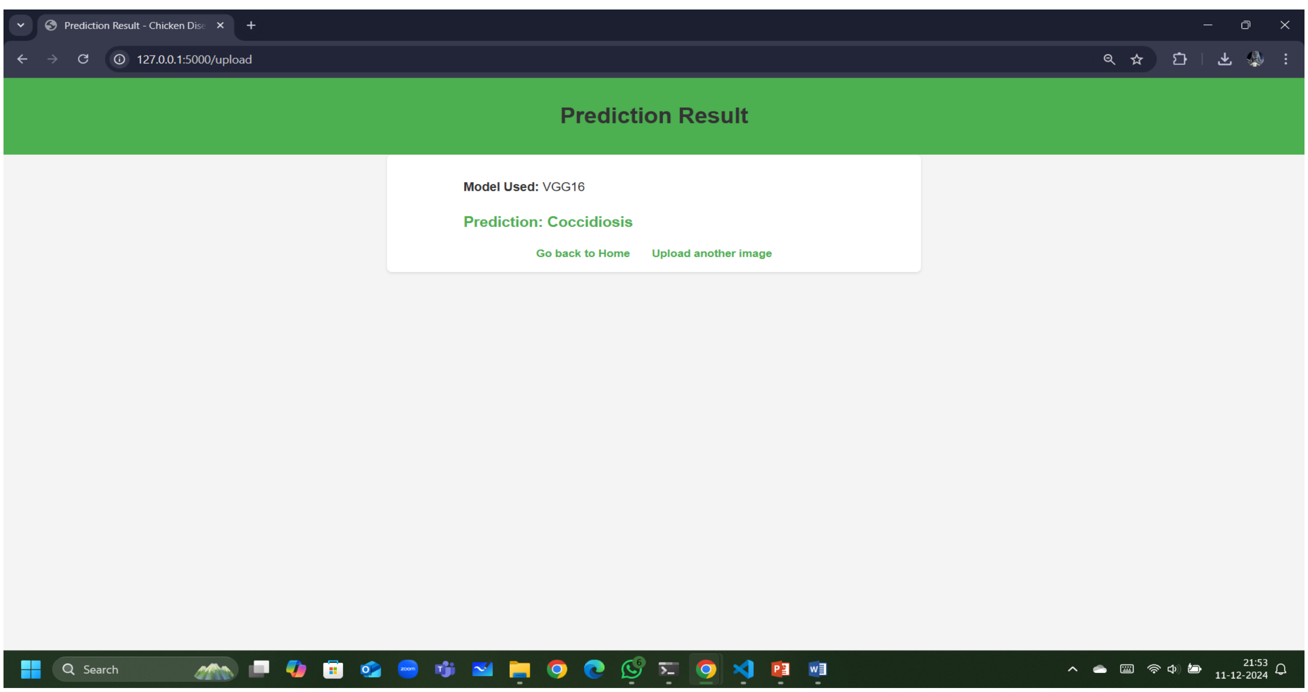

fig 6.4 :VGG16 Model Prediction and Detected Disease

Fig 6.4 Illustrates that The 4th page of dashboard where user can Identify the most accurate Disease without any Logical ,Technical glitches.

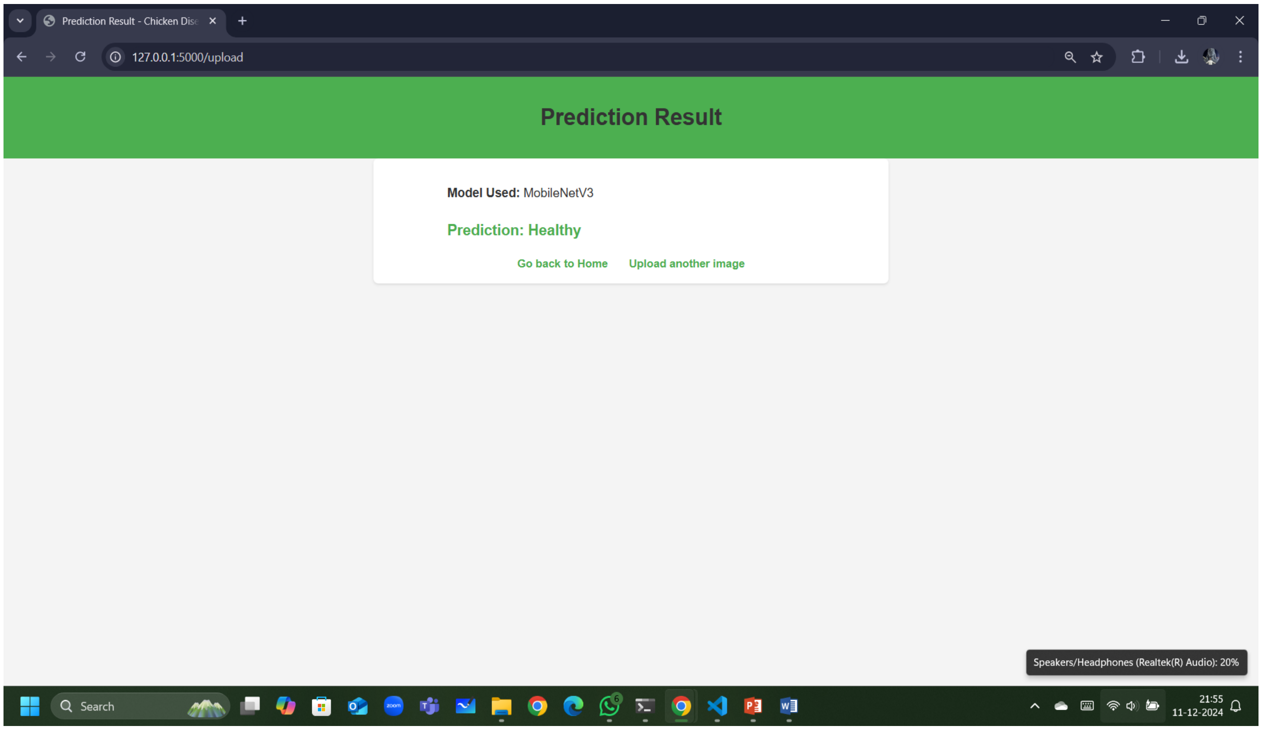

fig 6.5 :MobileNetV3 Model Prediction and Detected Disease

Fig 6.5 Illustrates that The output page of dashboard where user can Identify the most accurate Disease without any Logical ,Technical glitches by selecting MobileNetV3 ML Trained Model.

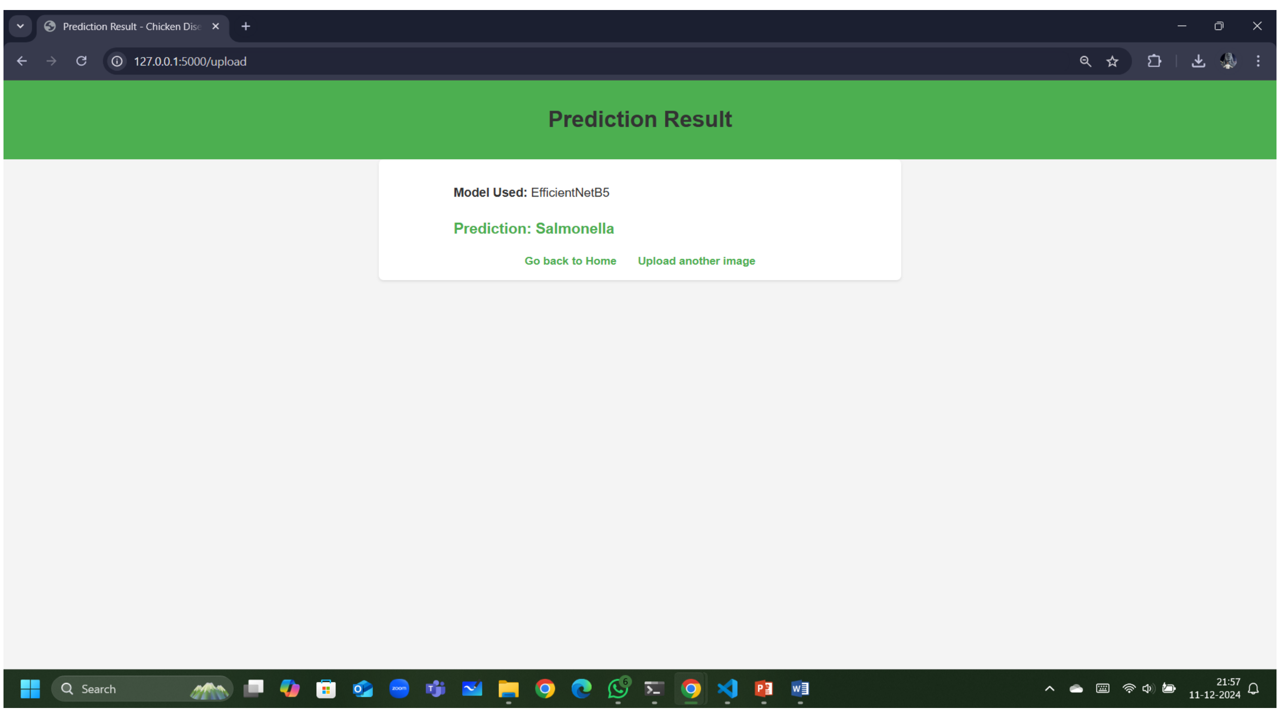

fig 6.6 :EfficientNetB5 Model Prediction and Detected Disease

Fig 6.6 Illustrates that The output page of dashboard where user can Identify the most accurate Disease without any Logical ,Technical glitches by choosing EfficientNetB5 ML Model.

CONCLUSION

In this project, we developed a robust and efficient system for the classification and detection of chicken diseases using Convolutional Neural Networks (CNN) and image classification techniques. By leveraging deep learning methods, our model was able to accurately diagnose a variety of common poultry diseases such as Newcastle disease, fowl pox, and coccidiosis from image data. The proposed system automates the traditionally labor-intensive process of disease identification, providing a scalable and time-efficient solution for poultry farmers and veterinarians.

Our results demonstrate that the CNN-based model significantly outperforms traditional diagnostic methods, achieving high accuracy and reliable performance in real-world settings. The system also offers the potential to enhance disease management practices by facilitating early detection, enabling timely treatment, and ultimately reducing economic losses in poultry farming. The deployment of this automated detection system can revolutionize poultry health monitoring, making it accessible to a wider range of farmers, particularly in large-scale operations and remote areas.

Future work will focus on improving the model's generalization capabilities through continuous learning and expanding the dataset to include additional diseases and environmental variations. In conclusion, this project provides a critical step toward smarter, technology-driven poultry disease management, with the potential to improve food security and animal welfare through innovative, data-driven solutions.

FUTURE SCOPE OF THE PROJECT :

Integration with IoT and Smart Farming:Automated fecal imaging systems coupled with IoT devices can enable real-time monitoring and detection of diseases in large-scale poultry farms.

Advanced Machine Learning Models:Development of more sophisticated AI models, such as transformers and hybrid architectures, can improve classification accuracy.

Expanded Dataset Availability:Creation of standardized, large-scale fecal image datasets representing diverse chicken breeds, regions, and diseases will improve model robustness.

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IMPLEMENTATION OF MACHINE LEARNING - BASED FOWL DISEASE DETECTION USING CNN

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Abstract

Poultry husbandry is an essential contributor to global food security, yet it faces significant challenges from complaint outbreaks, leading to profitable losses and high mortality rates. This paper presents an automated funk complaint discovery Model using CNNs - Convolutional Neural Networks. By using image bracket ways, the system classifies fecal images into complaint orders similar as Newcastle complaint, coccidiosis, and healthy. The proposed result achieves high delicacy and scalability, offering growers a dependable tool for early complaint discovery and intervention. Experimental results demonstrate the effectiveness of models like VGG16, EfficientNetB5, and MobileNetV3, with EfficientNetB5 achieving superior performance

Keywords: Convolutional Neural Networks, Poultry variety diseases,VGG16 Layer ML model, and MobileNetV3 ML model, EfficientNetB5, Deep Learning.

1. Preface

Flesh husbandry is vital for addressing the global demand for beast protein. still, conditions like coccidiosis, Newcastle complaint, and Salmonella pose significant pitfalls to productivity. Traditional individual styles calculate on homemade examination, which is time- ferocious and error-prone. Advances in deep literacy have enabled automated systems able of assaying large datasets to identify complaint patterns efficiently. This paper proposes a CNN- grounded system for funk complaint discovery using fecal image datasets, furnishing an accessible and accurate result for growers. Flesh husbandry is vital for addressing the global demand for beast protein. still, conditions like coccidiosis, Newcastle complaint, and Salmonella pose significant pitfalls to productivity. Traditional individual styles calculate on homemade examination, which is time- ferocious and error-prone. Advances in deep literacy have enabled automated systems able of assaying large datasets to identify complaint patterns efficiently. This paper proposes a CNN- grounded system for funk complaint discovery using fecal image datasets, furnishing an accessible and accurate result for growers.

2. Literature Review

colorful studies have explored complaint discovery in flesh using machine literacy. For case, Banakar et al.(2021) employed sound features to diagnose avian conditions with 91.15 delicacy. Another study by Vandana et al.(2023) employed the EfficientNetB7 model for fecal image analysis, achieving 97.07 delicacy. These styles demonstrate the eventuality of noninvasive ways for complaint discovery but highlight challenges similar as dataset imbalance and real- time deployment.

3. Methodology

3.1 Dataset Preparation

The dataset comprises labeled funk fecal images distributed as" Healthy,"" Coccidiosis,"" Newcastle Disease," and" Salmonella." Images were resized to 224x224 pixels for preprocessing. Data addition ways similar as gyration, flipping, and discrepancy adaptation were employed to ameliorate model conception.

3.2 Model Infrastructures

1. **VGG16 :** A deep CNN with 16 layers, primarily used for point birth through its invariant armature.
2. **EfficientNetB5 :** An optimized CNN exercising emulsion scaling to balance depth, range, and resolution.
3. **MobileNetV3 :** A featherlight CNN designed for mobile and edge bias, combining effectiveness with high performance.

3.3 Training Process

The models were trained using TensorFlow with categoricalcross-entropy as the loss function and Adam optimizer. The dataset was resolve into 70 training, 10 confirmation, and 20 testing subsets. Beforehand stopping and learning rate decay were applied to help overfitting.

4. Results and Discussion

The performance of the models was evaluated using metrics such as accuracy, precision, recall, and F1-score.Comparison Analysis performed on all Three Machine Learning Models and achieved good Accuracy by EfficientNetB5 Model.

ML Model Name	Accuray (%)	Precision (%)	Recall (%)	F1-Score (%)
VGG16	89.2	88.5	87.8	88.1
EfficientNetB5	96.7	96.9	96.4	96.6
MobileNetV3	93.4	93.2	92.8	93.0

EfficientNetB5 outperformed other models in terms of delicacy and robustness, making it suitable for real- world operations.

5. Perpetration

The system was stationed as a web operation using Beaker. The interface allows druggies to upload fecal images and choose a preferred model for vaticination. The backend processes images through the trained CNN model and returns the prognosticated complaint order. The operation is designed to be stoner-friendly, enabling growers to pierce individual results without specialized moxie.

6. Confusion Matrix

The confusion matrix states an analysis of the bracket model’s performance of EfficentNet-B5 across four orders Healthy(Class 0), Coccidiosis(Class 1), Newcastle Disease(Class 2), and Salmonella(Class 3). The model shows high delicacy with true positive counts of 243 for Healthy, 226 for Coccidiosis, 53 for Newcastle Disease, and 255 for Salmonella. Misclassifica-tions are minimum, similar as 5 Healthy samples labeled as Coccidiosis and 1 NewcastleDisease case misclassified as Healthy.

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Class 3(Salmonella) demonstrates the stylish performance with nearly no crimes. While the results validate the model's trustability in relating conditions, slight misclassifications in Classes 1 and 2 indicate room for enhancement through ways like fresh data collection or fine- tuning. These findings accentuate the effectiveness of the EfficientNetB5 model in flesh complaint bracket.The confusion matrix provides an analysis of the bracket model's performance of EfficentNet- B5 across four orders Healthy(Class 0), Coccidiosis(Class 1), Newcastle Disease(Class 2), and Salmonella(Class 3). The model shows high delicacy with true positive counts of 243 for Healthy, 226 for Coccidiosis, 53 for Newcastle Disease, and 255 for Salmonella. Misclassifications are minimum, similar as 5 Healthy samples labeled as Coccidiosis and 1 Newcastle Disease case misclassified as Healthy. Class 3(Salmonella) demonstrates the stylish performance with nearly no crimes, slight misclassifications in Classes 1 and 2 indicate room for enhancement through ways like fresh data collection or fine- tuning. These findings accentuate the effectiveness of the EfficientNetB5 model in flesh complaint bracket.

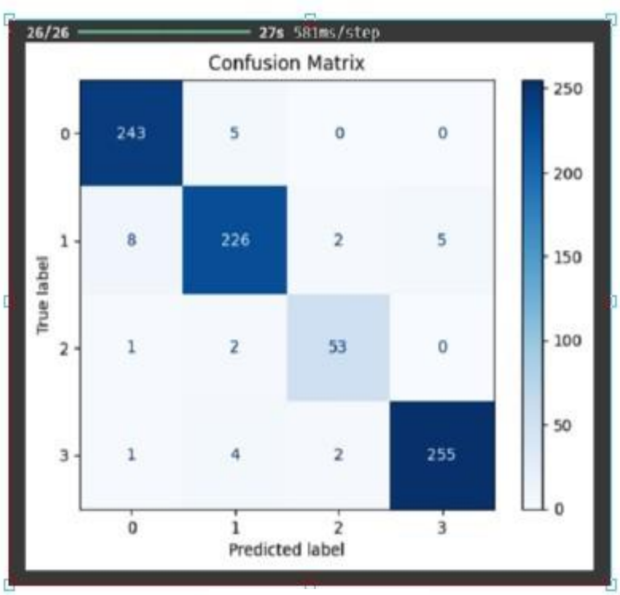

fig 6.1 : Matrix Representation of EfficientNetB5

7. Analysis Graph

The graphs illustrate the training and Testing accuracy and precision of the model over 20 epochs. The accuracy plot shows that the model achieves rapid improvement in both training and Evaluation accuracy within the first few epochs, stabilizing around 95.4% for training accuracy and 96.2% for validation accuracy, demonstrating effective learning. The loss graph reflects a sharp decline in both training and Testing loss during the initial epochs, with training loss converging close to zero and validation loss stabilizing at a low value of 0.1697. These trends indicate that the model generalizes well, with no signs of significant overfitting or underfitting, confirming its reliability for disease detection tasks.

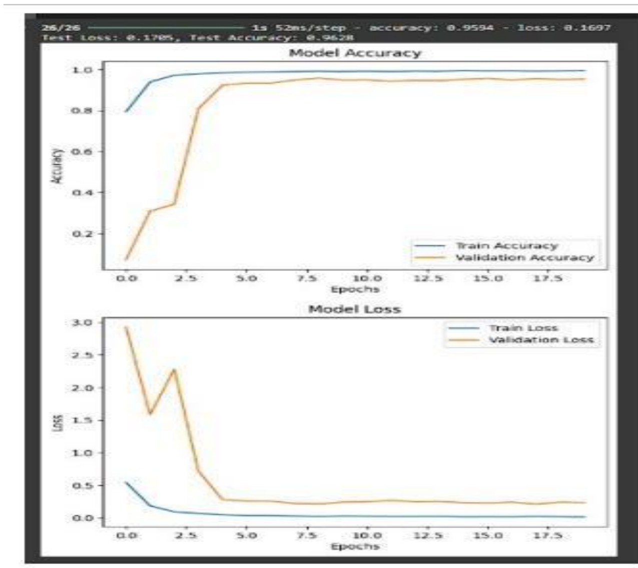

fig 7.1 : Trained model Accuracy Graph

8. Conclusion and Unborn Compass

This study demonstrates the eventuality of CNNs in automating funk complaint discovery. The EfficientNetB5 model, in particular, provides high delicacy and with vertical and Horizontal Types of scalability, offering a feasible result for flesh health operation. unborn work will concentrate on expanding the dataset to include different envir-onmental conditions and integrating IoT bias for real- time monitoring .This study demonstrates the eventuality of CNNs in automating funk complaint discovery. The EfficientNetB5 model, in particular it provides high delicacy and scalability, offering a feasible result for flesh health operation. unborn work will concentrate on expanding the dataset to include different environmental conditions and integrating IoT bias for real- time monitoring.

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